KAIJIE LIU

liu6kx@dukes.jmu.edu | (434) 327 0273 | Crozet, VA

EDUCATION

James Madison University (JMU), USA **2024-2028 anticipated**

Ph.D. in Assessment and Measurement; *Advisor: Dr. John D. Hathcoat*

University of Virginia (UVA), USA **2022-2023**

M.Ed. in Quantitative Analytics in Education and the Social Sciences; *Advisor: Dr. Vivian C. Wong*

University of Konstanz (UKN), Germany **2017-2019**

M.Sc. in Economics; *Advisor: Dr. Ralf Brueggemann*

Thesis: International Connectedness of Economic Variables – Evidence from New Time Series Methods

South China University of Technology (SCUT), China **2006-2010**

Bachelor of Economics; *Advisor: Dr. Min Li*

Thesis: Influences of Innovation on the Competitiveness of Xiqiao's Textile Industry Cluster

PROFESSIONAL EXPERIENCE

Doctoral Assistant

Student Success Analytics, JMU **Aug 2025-Present**

- Manage and clean large datasets from PeopleSoft by identifying missing values and duplicates, creating numeric and categorical labels, and merging data for analysis.
- Conduct statistical analyses using propensity score methods and confirmatory factor analysis to assess the impact of campus campaigns and programs on improving student engagement and experience.

Center for Assessment and Research Studies (CARS), JMU

Data Management | Assessment Day Team **Jun 2024-Present**

- Lead data analysis projects on JMU students' political engagement, performing cross-cohort comparisons and senior-versus-freshmen analyses, delivering comprehensive reports to the James Madison Center for Civic Engagement.
- Collaborate with team members on data collection and management for the university-wide Assessment Day.
- Conduct quality assurance checks on assessments, data management syntax, and assessment-related documentation to ensure accuracy and reliability.
- Communicate with faculty and university offices to address and accommodate student testing needs.
- Contribute to a research project examining test performance, test-taking efforts, and motivations across varied test orderings.
- Process and combine data for deliveries of requested research-ready datasets.

CARS General Team **Jun 2024-May 2025**

- Analysed student assessment data for the JMU General Education Program.
- Assisted in maintaining a program tracking system using Smartsheet.
- Reviewed literature for measures regarding Artificial Intelligence (AI) literacy in higher education.
- Managed projects related to AI applications in assessment, overseeing goals, timelines, and deliverables.
- Planned assessment-related events on campus, preparing materials and coordinating logistics.

Graduate Intern | Data Specialist **Feb 2024-Jun 2024**

Instructional Services Department, Fairfax County Public Schools

- Prepared and merged student achievement and demographic datasets for analysis and departmental reporting.
- Conducted quantitative and qualitative analyses of *"Portrait of a Graduate - Presentations of Learning"* data.
- Evaluated summer learning programs for elementary and middle school students using statistical methods to measure academic growth and program effectiveness.
- Performed quality assurance checks on multi-year school information datasets to ensure accuracy and consistency for ongoing research.

Master's Internship **Jun 2023-Sep 2023**

Collaboratory Replication Lab, School of Education and Human Development, UVA

- Conducted descriptive and exploratory analyses for a large-scale national teacher survey on science instruction for elementary students with learning disabilities.
- Compiled multi-level datasets by merging participant-level survey data with school- and district-level information from the National Center for Education Statistics and the Stanford Education Data Archive datasets.
- Designed and built data collection tools for a classroom observational study, contributing to the data infrastructure for the Special Education Research Accelerator project.
- Developed text message survey distribution procedures and authored a detailed procedural manual.
- Assisted in the design of a data tracking system using Smartsheet and created flowcharts to visualize workflows for operational efficiency.
- Established variable naming conventions, created consort diagrams, and developed data architecture for research projects focused on special education and science instruction.

Graduate Research Assistant **Oct 2022-May 2023**

Collaboratory Replication Lab, School of Education and Human Development, UVA

- Contributed to item development for a national teacher survey. Built and programmed survey flows, troubleshooting, piloting, and refining projects on survey platforms.
- Assisted in drafting protocols for research partner and teacher recruitments.
- Led survey distribution to a U.S. representative sample of 3,700+ participants across 400 schools, implementing an embedded randomized experimental design. Monitored response rates and deployed reminders to maximize participation.
- Prepared research-ready datasets with detailed documentation, including data cleaning, variable computing, and missing/erroneous data resolving.
- Analysed research partner characteristic data, creating tables, charts, and graphs to visualize results, and drafted subsections for the project annual report.
- Developed codebooks and standardized data management syntax for a teacher preparation project.

Assistant Treasurer **2012-2014**

Jin Kai Sheng Group Company, Guangdong, China

Project Associate **2011-2012**

Jin Kai Sheng Group Company, Guangdong, China

Customer Manager **2010-2011**

China Galaxy Securities Company

RESEARCH EXPERIENCE

Second-Year Research Project (IRB-FY26-591) **Oct 2025-Present**

Department of Graduate Psychology, JMU

- Develop a shorter version of the Generative AI Literacy Assessment Test (GLAT), referred to as Mini-GLAT.
- Analyze GLAT data collected from multiple student groups during the 25-26 school year, including first-year, mid-career, and senior students.

- Apply classical test theory and item response theory methods in conducting item-level analysis.
- Examine differential item functioning (DIF) across student groups.
- Diagnose and identify problematic items; reserve a smaller set of items to construct Mini-GLAT.
- Cross-validate Mini-GLAT using psychometric measurement methods.

First-Year Research Project (IRB-FY25-399) **Aug 2024-May 2025**

Department of Graduate Psychology, JMU

- Investigated the use of generative AI as an ethical reasoning essay rater, applying a rubric from JMU's Ethical Reasoning in Action program.
- Developed and refined prompt engineering strategies to employ Microsoft Copilot as a rater and evaluated inter-rater agreement between AI and human raters.
- Applied a many-facet Rasch model to examine (i) the severity/leniency of Copilot and (ii) its effectiveness in differentiating rubric categories compared to human raters.

Independent Study, Master's Thesis **Apr 2019-Aug 2019**

Department of Economics, UKN

- Utilized daily stock market indices data from eight markets spanning from 2006 to 2019.
- Measured international connectedness in financial markets using the out-of-sample forecast error variance decomposition (FEVD) approach within a generalized vector autoregression model.
- Conducted a comparative analysis with the in-sample FEVD approach, a common method in the network literature.

Independent Study, Diploma Thesis **Feb 2010-Jun 2010**

School of Economics and Finance, SCUT

- Investigated the competitive advantages of industry innovation in Xiqiao's textile industry cluster.
- Administered questionnaires to 100+ local business owners to assess the influences of innovation and evaluate clustering effects.
- Applied the hierarchical decomposition method and the analytic hierarchy process for data analysis, drawing insights from Porter's Diamond Model for context.

Undergrad Research Assistantship **Mar 2008-Jul 2008**

School of Economics and Finance, SCUT (Supervisor: Dr. Qi Xiao)

- Conducted research on the subprime mortgage crisis, including data and material collection.
- Reviewed literature on topics such as Behavioral Finance Theory, Liquidity, and Financial Supervision.
- Participated in discussions regarding the mechanism of the financial crisis and contributed to reports drafting.

CONFERENCE PRESENTATIONS

Liu, K., Kush, J. M. (2026, April). *Propensity Score Weighting and Effect Estimation with Sampling Weights and Plausible Values.* Paper presented at the National Council on Measurement in Education (NCME) Annual Meeting, Graduate Student eBoards, Conference, Los Angeles, CA.

Liu, K., Hathcoat, J.D., Bao, Y. & Castleton, H.R. (2026, April). *Can GenAI score ethical reasoning essays? Evaluating Copilot with interrater agreement and Rasch modeling.* Paper presented at the American Educational Research Association (AERA) Annual Meeting, Consortium of State and Regional Educational Research Associations (SRERA), Distinguished Paper Session #3, Conference, Los Angeles, CA.

Liu, K., Hathcoat, J.D., Bao, Y. & Castleton, H.R. (2025, October 22-24). *Can GenAI score ethical reasoning essays? Evaluating Copilot with interrater agreement and Rasch modeling.* Paper presented at the Northeastern Educational Research Association (NERA) Annual Conference, Conference, Trumbull, CT.

Henriques, J., Pastor D., **Liu, K.**, Castleton H.R., Gilmore J. (2025, October 22-24). *Optimizing Test Sequencing: The Effects of Assessment Order on Student Effort and Performance.* Paper presented at the Northeastern Educational Research Association (NERA) Annual Conference, Conference, Trumbull, CT.

PUBLICATIONS

Liu, K. (2025). Review of "Relationships between structural stigma, societal stigma, and minority stress among gender minority people." In S. J. Finney (Ed.), *Structural Equation Modeling for Good*. Pressbooks. https://pressbooks.lib.jmu.edu/semforgood/

Manuscripts Under Review

Liu, K., Hathcoat, J.D., Bao, Y. & Castleton, H.R. (Under review, 2025). Evaluating Copilot as an Essay Rater: Interrater Agreement and Rasch Analysis of Ethical Reasoning Scores.

HONORS & AWARDS

Lorne H. Woollatt Distinguished Paper Award, Northeastern Educational Research Association 2025

Graduate Travel Grant Award, Office of Student Awards, Initiatives and Research, James Madison University 2025

PROFESSIONAL AFFILIATIONS

National Council on Measurement in Education

American Educational Research Association, Division D

Northeastern Educational Research Association

RELEVANT COURSE WORK

Data Management and Analysis

Quantitative Methods II

Measurement Theory

Structural Equation Modeling

Rasch Analysis

Design and Analysis of Field Experiments

Scholarly Communication

Independent Study: Intro Python & Machine Learning

Foundations of Educational Research

Multivariate Statistics in Psychology

Assessment Methods & Instrument Design

Quasi-Experimental Research

Multilevel Modeling

Generalized Linear Models

Stats Consultation

SKILLS

Statistical Software: R, Python, SAS, Stata, SPSS, WINSTEPS, JMulTi, MATLAB

Experience: Qualtrics, REDCap, SQL, Tableau, Smartsheet, Mailchimp, Lucidchart, Microsoft Office, LaTeX

LANGUAGES

Chinese (*Native*); English (*Fluent*); German (*Intermediate*).